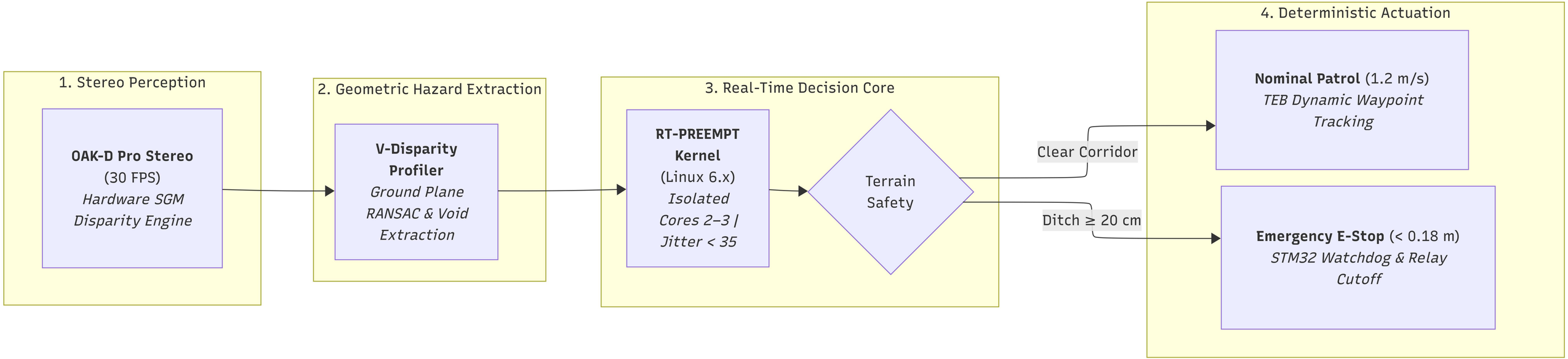
1. Stereo Perception
OAK-D Pro Stereo
(30 FPS)
Hardware SGM
Disparity Engine
2. Geometric Hazard Extraction
V-Disparity
Profiler
Ground Plane
RANSAC & Void
Extraction
3. Real-Time Decision Core
RT-PREEMPT
Kernel
(Linux 6.x)
Isolated
Cores 2–3 |
Jitter < 35
Terrain
Safety
Clear Corridor
Ditch ≥ 20 cm
4. Deterministic Actuation
Nominal Patrol (1.2 m/s)
TEB Dynamic Waypoint
Tracking
Emergency E-Stop (< 0.18 m)
STM32 Watchdog & Relay
Cutoff